

Drugs for Diabetes

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Conflict of Interest Statement

The presenters do not have a financial interest, arrangement or affiliation with any organization that could be perceived as a real or apparent conflict of interest in the context of this presentation.

Learning Objectives

1. Describe the pharmacology of insulin preparations with respect to onset and duration of action, listing the primary preparation(s) in each group.
2. Describe the mechanisms of action, use, and adverse effects of hypoglycemic agents for the treatment of diabetes mellitus: sulfonylureas, biguanides, thiazolidinediones, alpha-glucosidase inhibitors, GLP-1 agonists, amylin agonist, DPP-4 inhibitors, SGLT2 inhibitors, glinides.
3. Describe underlying co-morbidities that impact the choice of hypoglycemic agents.

Diagnosis of Diabetes (ADA)

A hemoglobin A1c (HbA1c) level of 6.5% or higher; **or**

A fasting plasma glucose (FPG) level of 126 mg/dL (7 mmol/L) or higher; fasting is defined as no caloric intake for at least 8 hours, **or**

A 2-hour plasma glucose level of 200 mg/dL (11.1 mmol/L) or higher during a 75-g oral glucose tolerance test (OGTT), **or**

A random plasma glucose of 200 mg/dL (11.1 mmol/L) or higher in a patient with classic symptoms of hyperglycemia (i.e., polyuria, polydipsia, polyphagia, weight loss) or hyperglycemic crisis

Diabetes Mellitus

Chronic disease characterized by hyperglycemia and glucosuria.

Type 1 (T1D)

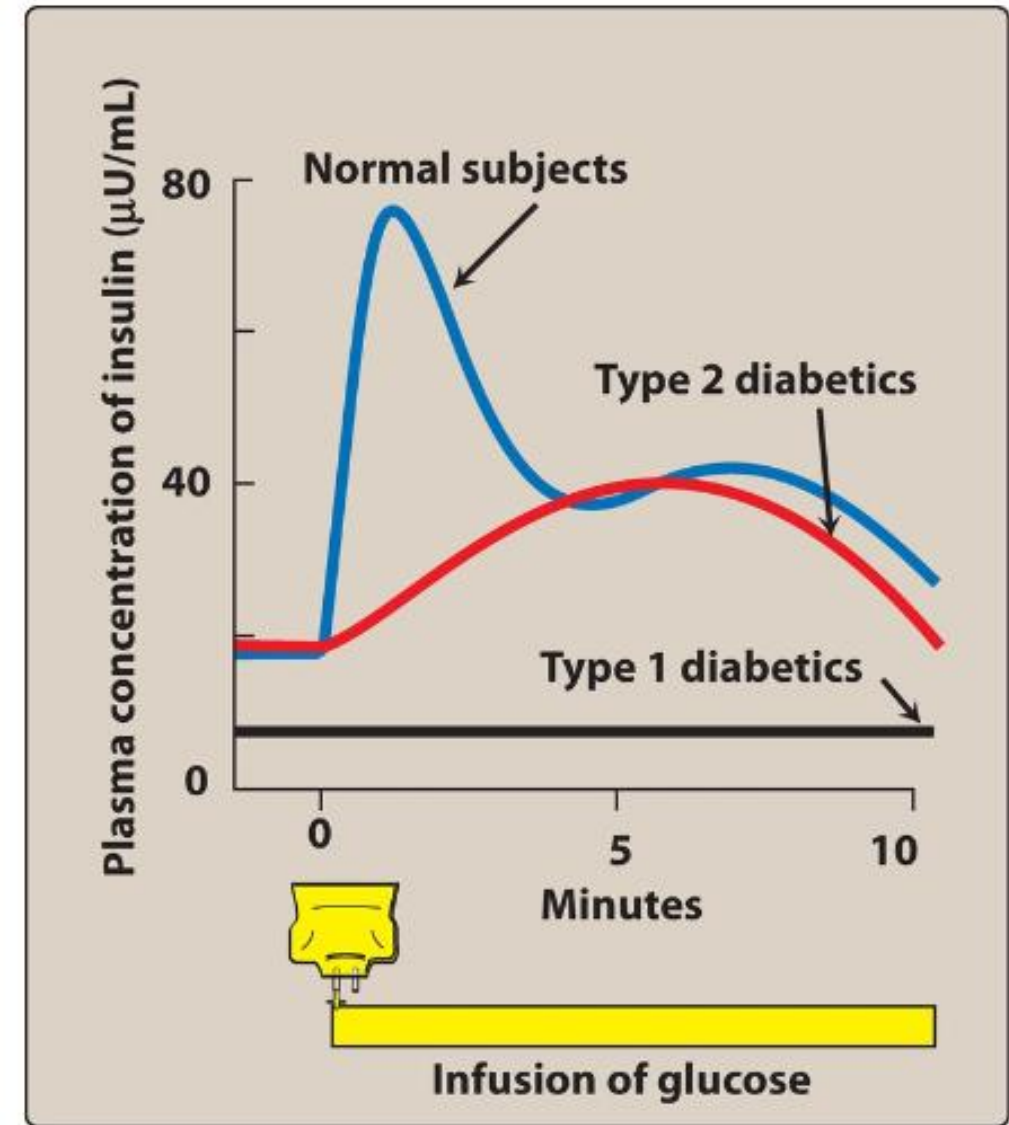
Patient secretes little to no insulin (absolute deficiency).
Autoimmune disorder.

Type 2 (T2D)

Most common type (90%).

Insulin resistance and relative insulin deficiency.

Associated with obesity, dyslipidemia, hypertension, and other conditions.



Source: Lippincott's Illustrated Review of Pharmacology, 7th Ed.

DM Characteristics

	Type 1	Type 2
Onset	Most often abrupt	Gradual
Clinical Symptoms	Polydipsia, polyphagia, polyuria	Often none
Ketosis	Frequent	Infrequent (except with infection or stress)
Endogenous Insulin	Negligible	Present but often ineffective
Insulin Therapy	Immediate	Years after diagnosis
Non-Insulin Hypoglycemic Agents	Rarely used	Clinically indicated

Drug-Induced Causes of Type 2 Diabetes

Glucocorticoids

Thyroid hormones

β -blockers

Post-transplant drugs

- (tacrolimus, cyclosporine, sirolimus)

Thiazides

Anti-retrovirals

Anti-psychotics

Statins

Nicotinic acid



Case

A 19 yo woman with Type 1 DM on insulin presents to your clinic for routine follow up. She was diagnosed with Type 1 DM 3 months ago, and she is currently taking insulin glargine 32 units at nighttime, and insulin aspart 6 units with breakfast, 6 units with lunch and 8 units with dinner. She denies any sweats, confusion, jitteriness. She brings her meter with her today, which shows fasting AM blood sugars of 170-190, with post-prandial sugars of 120-150.

What is the glucose target goal for this patient?

ADA Recommended Goals of Therapy

Test	Goals
HgA1c	<7% (checked every 3 months)
Preprandial plasma glucose	80-130 mg/dl
Peak Postprandial plasma glucose	<180 mg/dl
Blood pressure	<140/90 mmHg
Lipid	Check every 1-2 years
Eye	Yearly exam
Feet	Every visit exam
Urinary microalbumin	Yearly

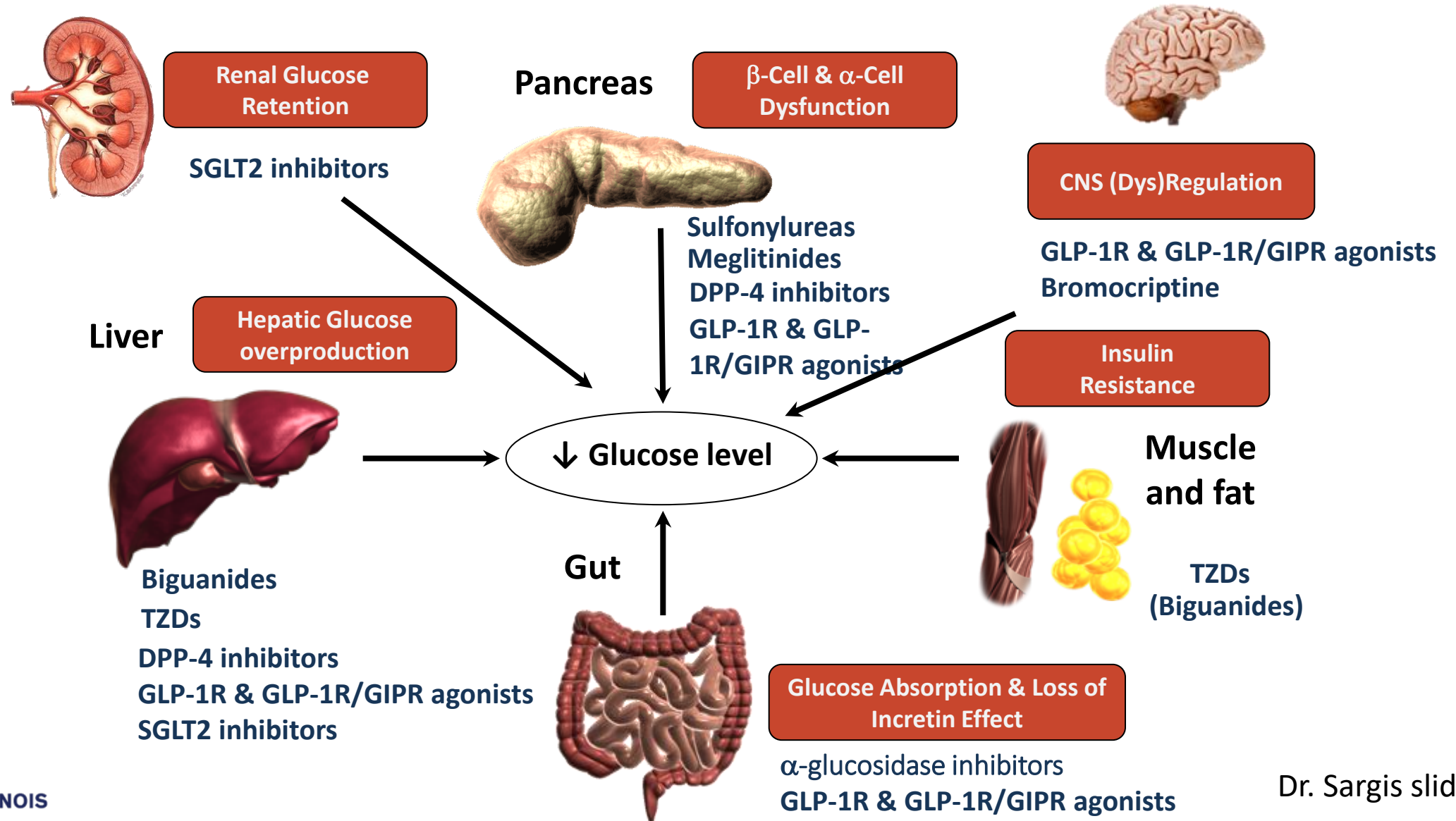
Let's talk about Drugs!

Treatment of type 1 diabetes will involve one or more forms of insulin while regimens for type 2 diabetes include monotherapy and combinations of two or three drugs from different classes.

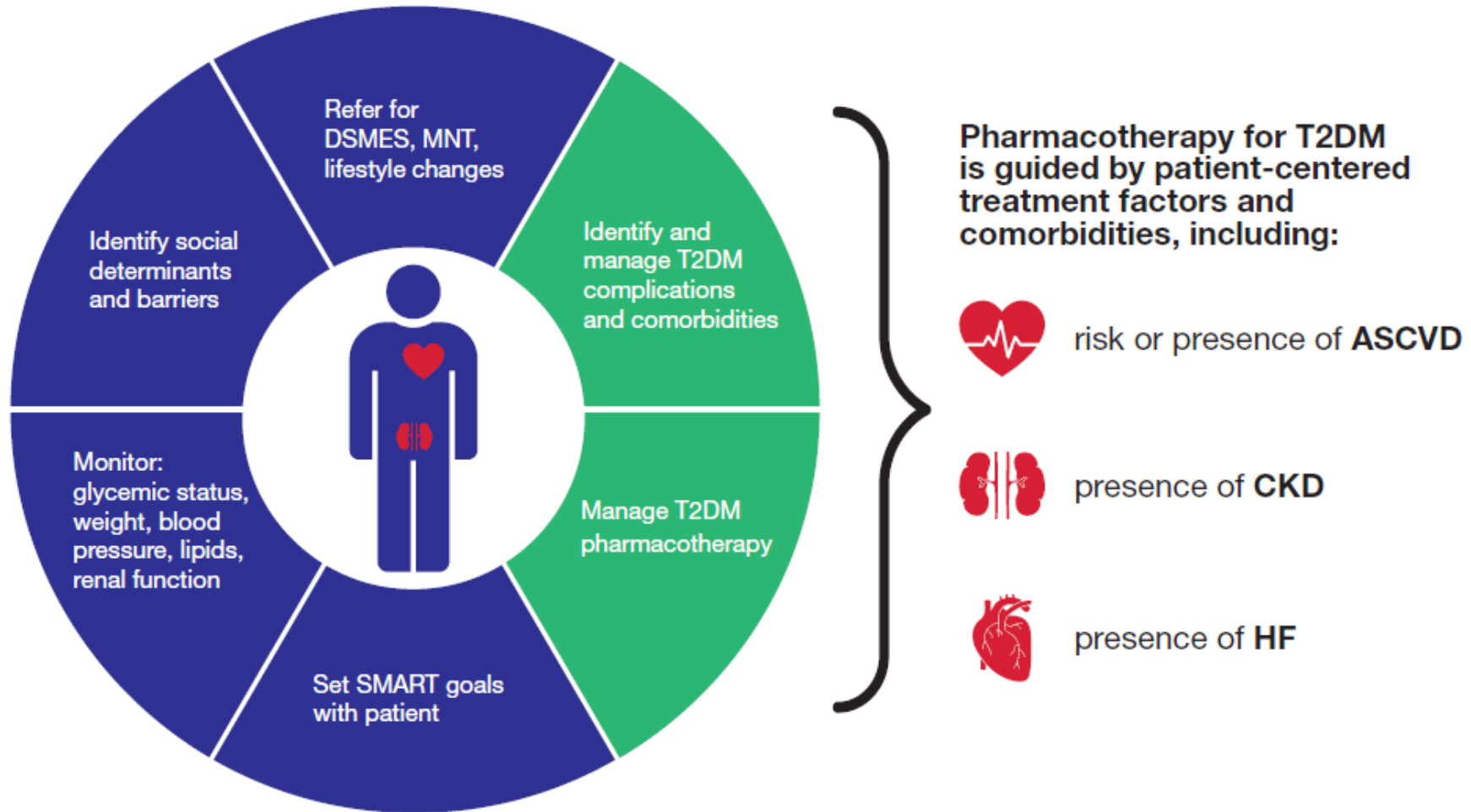
- Classes of antidiabetic drugs:
 - Insulins
 - Biguanides
 - Thiazolidinediones
 - Dipeptidyl peptidase-4 (DPP-4) inhibitors
 - Sodium Glucose Co-Transporter 2 (SGLT2) Inhibitors
 - Glucagon-like peptide-1 (GLP-1) receptor agonists and glucose-dependent insulintropic polypeptide (GIP) receptor/GLP-1R dual agonists
 - Sulfonylureas
 - Meglitinides*
 - Amylin analogue (pramlintide)*
 - α -glucosidase inhibitors (acarbose)*
 - Dopamine agonists (bromocriptine)*
 - Bile acid sequestrants (colesevelam)*

*Infrequently used but FDA-approved.

Target Sites of Anti-Diabetic Agents



Which agent to start --- think about the patient's co-morbidities



ASCVD coronary artery disease, cerebrovascular disease, or peripheral arterial disease, CKD eGFR < 60 ml/min/1.73 m² or albumin-to-creatinine ratio (ACR) ≥ 30 mg/g, HF reduced LVEF < 40% (HFrEF), mildly reduced LVEF 40-49% (HFmrEF), and preserved LVEF > 50% (HFpEF), DSMES diabetes self-management education and support, MNT medical nutrition therapy, SMART specific measurable achievable relevant time-bound

	Drug category	Drug names	Mechanism	Strengths/ uses	Adverse effects
Drugs that reduce insulin resistance	Biguanides	Metformin	Stimulate GLUT4, ↓ gluconeogenesis, ↓ gut glucose reabsorption	First-choice drug	Lactic acidosis (rare), GI upset
	Thiazolidinediones	Rosiglitazone, pioglitazone	↑PPAR-γ, causing cells to store fatty acids and take up glucose	Can't take metformin	Fluid retention, bladder cancer
Drugs that stimulate insulin release	GLP-1 agonists	Exenatide, liraglutide, dulaglutide	Bind to GLP-1 receptor, ↓ gastric emptying	Cardiovascular disease, weight loss	GI upset, acute pancreatitis
	DPP-4 inhibitors	Sitagliptin, linagliptin, saxagliptin	Inhibit DPP-4, increase GLP-1	Weight neutral	Pancreatitis
	Sulfonylureas	Glipizide, glyburide	Depolarize β cells, causing calcium influx and ↑ insulin release	Inexpensive, time-tested	Hypoglycemia, weight gain
	Meglitinides	Repaglinide, nateglinide			Hypoglycemia, weight gain

Drugs that increase renal glucose excretion	SGLT2 inhibitors	Canagliflozin, dapagliflozin, empagliflozin	↓ Proximal tubule glucose reabsorption	Cardio-vascular disease, weight loss	Dehydration, urinary tract infections, euglycemic diabetic ketoacidosis
Drugs that slow gastric emptying	Amylin analog	Pramlintide	↓ Glucagon secretion, ↓ gastric emptying, cause sense of satiety		GI upset
Drugs that inhibit GI carbohydrate absorption	α-Glucosidase inhibitors	Acarbose, miglitol	Inhibit α-glucosidase and pancreatic α-amylase		Flatulence

Question - Poll

Which of the following drugs is most likely to produce the largest decrease in hemoglobin A1c?

- A. pramlintide
- B. glyburide
- C. insulin
- D. metformin
- E. acarbose

Question – answer

Which of the following drugs is most likely to produce the largest decrease in hemoglobin A1c?

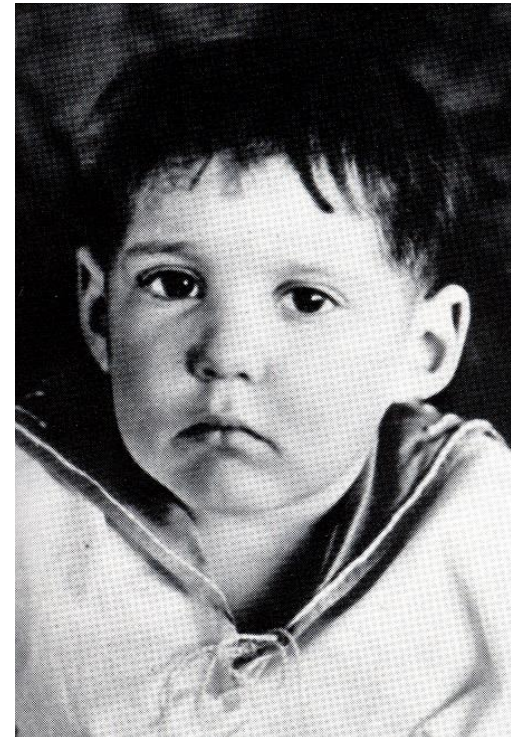
- A. pramlintide
- B. glyburide
- C. Insulin**
 - **>2% HgA1c reduction (if the blood sugar is very high, insulin is best)**
- D. metformin
- E. acarbose

Group 1: Insulin Therapy

The Miracle of Insulin

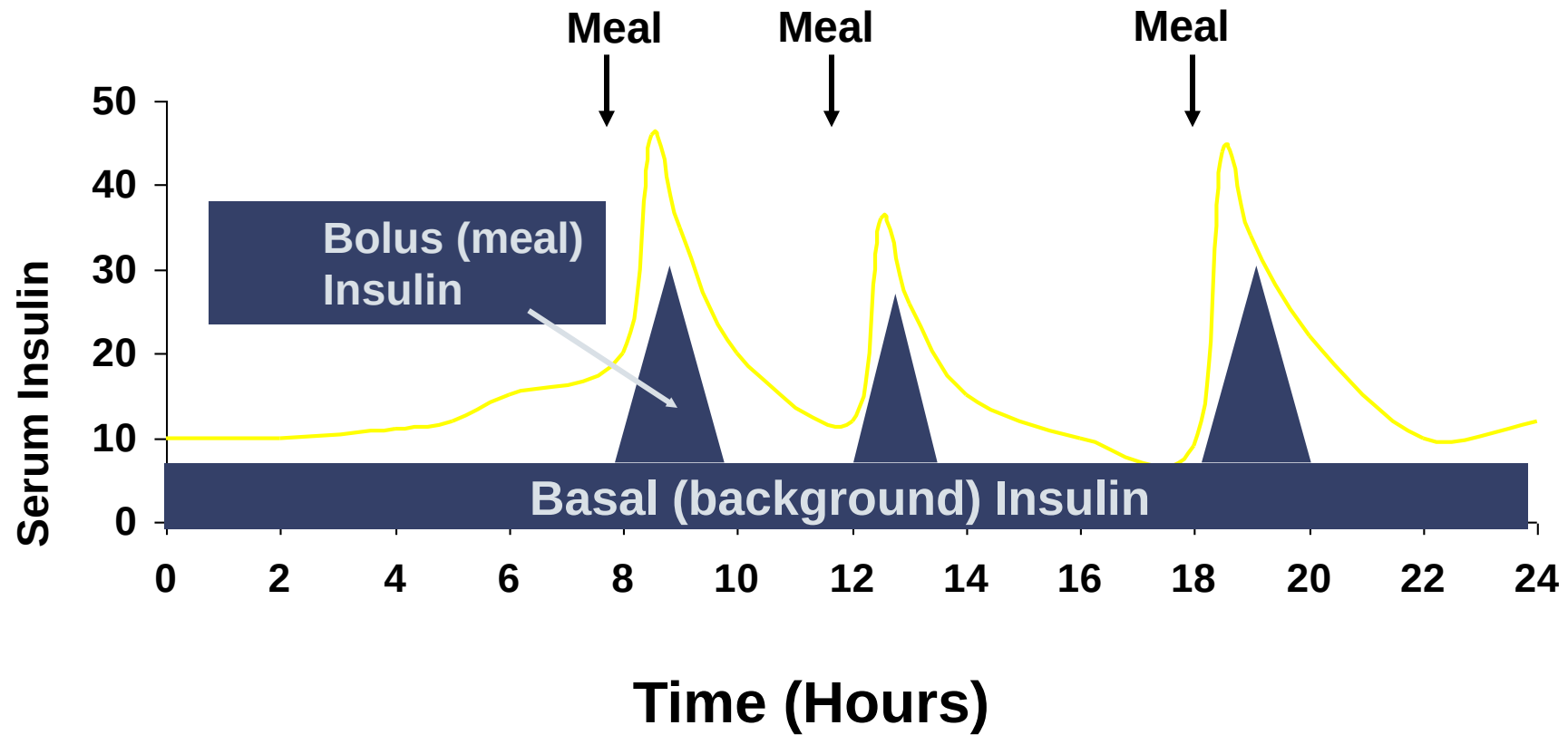


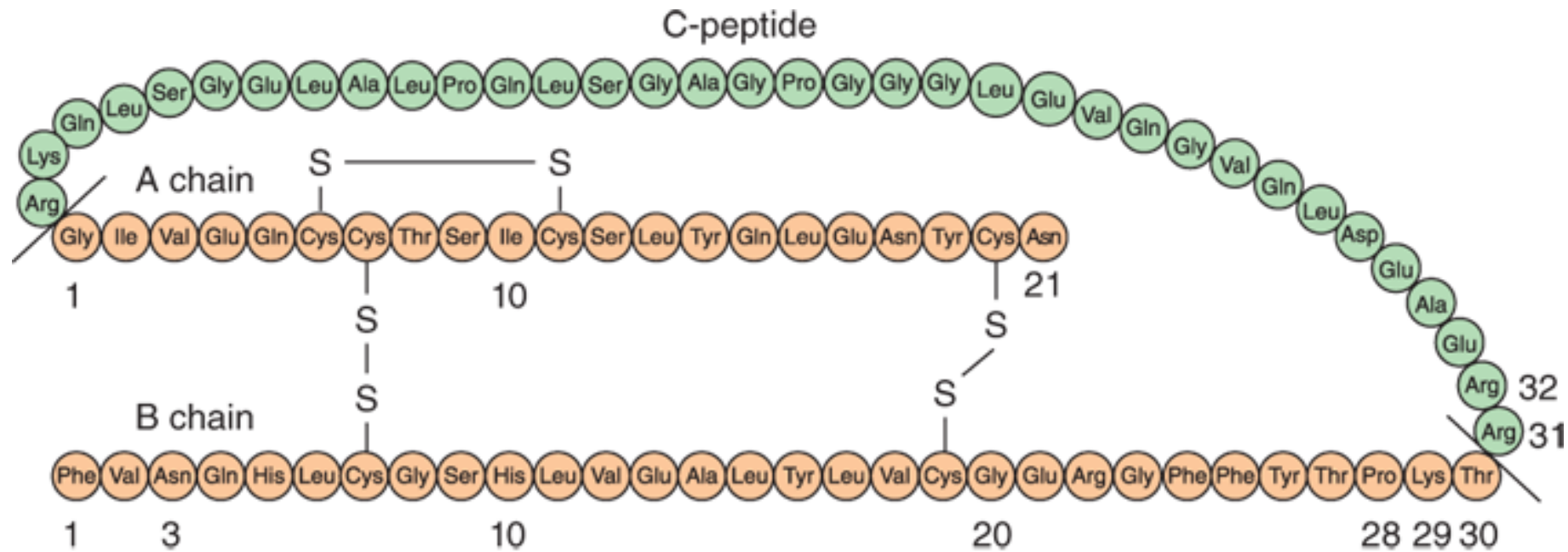
Patient J.L., December 15, 1922



Feb. 15, 1923

Normal Insulin Secretion





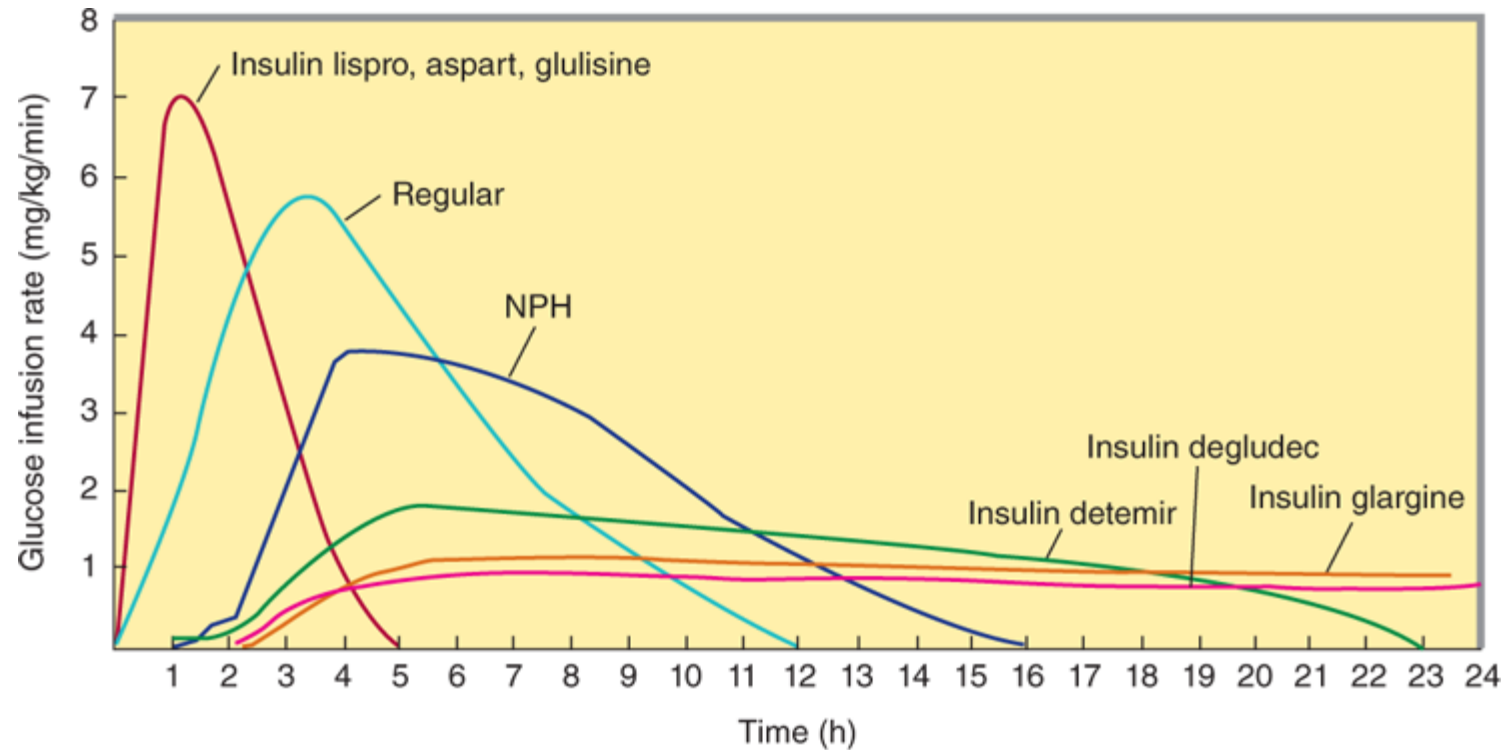
Source: Bertram G. Katzung:
Basic & Clinical Pharmacology, Fourteenth Edition
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Structure of human proinsulin (C-peptide plus A and B chains) and insulin. Insulin is shown as the shaded (orange color) peptide chains, A and B. Differences in the A and B chains

Pharmacology of Insulin

- Insulin preparations differ in: onset and duration of action
- Humulins: recombinant DNA synthesis of human insulin
- Insulin rapidly metabolized, not given orally
- Insulin preps classified as ultra-rapid-, rapid-, short-, intermediate- or long-acting
 - duration prolonged by conjugation with protein, crystallization with zinc, or AA substitutions
- Effect on Weight: Increase
- HbA1c: > 2% reduction (most effective)
- Side effects: **hypoglycemia, weight gain**, injection site discomfort

Pharmacokinetic Profiles of Insulins



Source: Bertram G. Katzung:
Basic & Clinical Pharmacology, Fourteenth Edition
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Extent and duration of action of various types of insulin as indicated by the glucose infusion rates (mg/kg/min) required to maintain a constant glucose concentration. The durations of action shown are typical of an average dose of 0.2–0.3 U/kg. The durations of regular and NPH insulin increase considerably when dosage is increased.

Insulins: Time Course of Action of Human Insulin Preparations

Insulin Preparation	Onset of Action (h)	Peak Action (h)	Effective Duration of Action (h)	Maximum Duration of Action (h)
Rapid acting analogs				
Lispro/Aspart/Glulisine	<0.25	0.5-1.5	2-4	4-6
Short acting				
Regular (soluble)	0.5-1	2-3	3-6	6-8
Regular U-500	0.5-1	2-3	12-24	>24
Intermediate acting				
NPH (isophane)	2-4	4-10	10-16	14-18
Long acting – basal analogs				
Glargine	1-2	None	>24	Unknown
Detemir	1-2	relatively flat	16-20	20-24
Degludec	1-2	None	28-36	>36
Inhaled Insulin				
Afrezza	<0.25	0.5-1	2-3	4-6

Insulins: Time Course of Action of Human Insulin Preparations

Most Combination products
AM doses affect pre lunch and
predinner sugars
PM doses affect bedtime and
AM fasting sugars

Insulin Preparation	Onset of Action (h)	Peak Action (h)	Effective Duration of Action (h)	Maximum Duration of Action (h)
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Combinations

70/30 (70% NPH, 30% regular)	0.5-1	Dual	10-16	14-18
Humalog 75/25	0.25	Dual	10-16	14-18
(75% NPL, 25% lispro)				
Humalog 50/50	0.25	Dual	10-16	14-18
(50% NPL, 50% lispro)				
Novolog 70/30	0.25	Dual	10-16	14-18
(70% NPA, 30% aspart)				
Ryzodeg 70/30	0.25	Dual	>24	>36
(70% degludec, 30% aspart)				
Fiasp U-100	0-0.05	0.5-1.0	2-4	2-4
(Insulin aspart combined with niacinamide for rapid absorption)				

Physiologic Multiple Injection Regimens: The Basal-Bolus Insulin Concept

Daily insulin need for a type 1DM with low insulin resistance –0.5 unit/kg/day

Basal insulin
~50-60% TDD



Bolus insulin
~40-50% TDD

- Controls glucose production between meals and overnight
- Near-constant levels

- Limits hyperglycemia after meals
- Immediate rise and sharp peak at 1 hour post-meal
- 10% to 20% of total daily insulin requirement at each meal

For ideal insulin replacement therapy, each component should come from a different insulin with a specific profile or via an insulin pump (with 1 insulin)

TDD, total daily dose.

Handelsman YH, et al. *Endocr Pract.* 2015;21(suppl 1):1-87.

The Basal/Bolus Insulin Concept

Basal Insulin

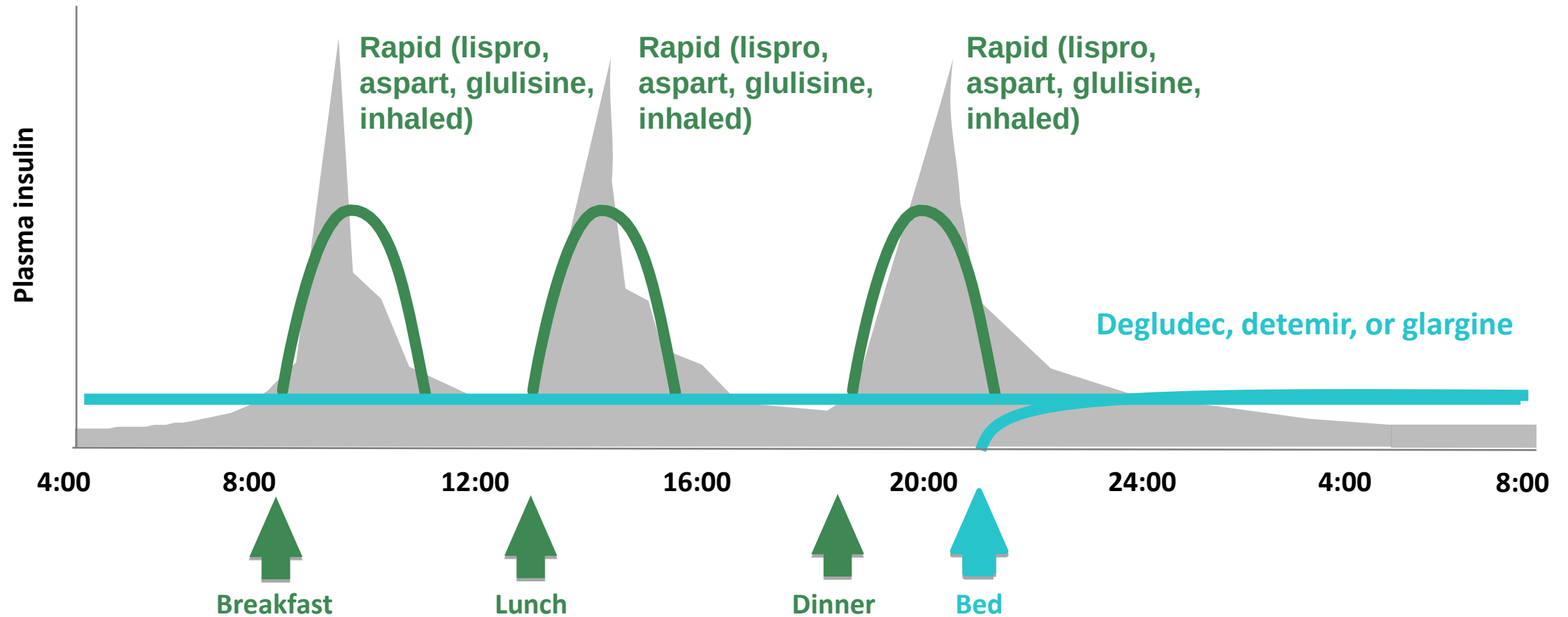
- **Suppresses glucose production between meals and overnight**
- Nearly constant levels
- 40 to 50% of daily needs

Bolus Insulin (Mealtime or Prandial)

- **Limits hyperglycemia after meals**
- Immediate rise and sharp peak at 1 hour
- 10% to 20% of total daily insulin requirement at each meal

Ideally, for insulin replacement therapy, each component should come from a different insulin with a specific profile...or is delivered with a system that allows for physiological matching.

Basal/Bolus Treatment



Question - Poll

A patient with long-standing type2DM is currently on 3 different oral hypoglycemic medications and on both basal and bolus insulins. Recent HgA1c continues to be suboptimal at 9.2%.

Which of the following insulins is the most appropriate to make adjustment for better glucose control?

A. Bolus insulin

B. Basal insulin

Question – answer

A patient with long-standing type2DM is currently on 3 different oral hypoglycemic medications and on both basal and bolus insulins. Recent HgA1c continues to be suboptimal at 10.9%.

Which of the following insulins is the most appropriate to make adjustment for better glucose control?

A. Bolus insulin

B. Basal insulin

Insulins: Time Course of Action of Human Insulin Preparations

Insulin Preparation	Onset of Action (h)	Peak Action (h)	Effective Duration of Action (h)	Maximum Duration of Action (h)
Rapid acting analogs				
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Insulin Delivery Systems

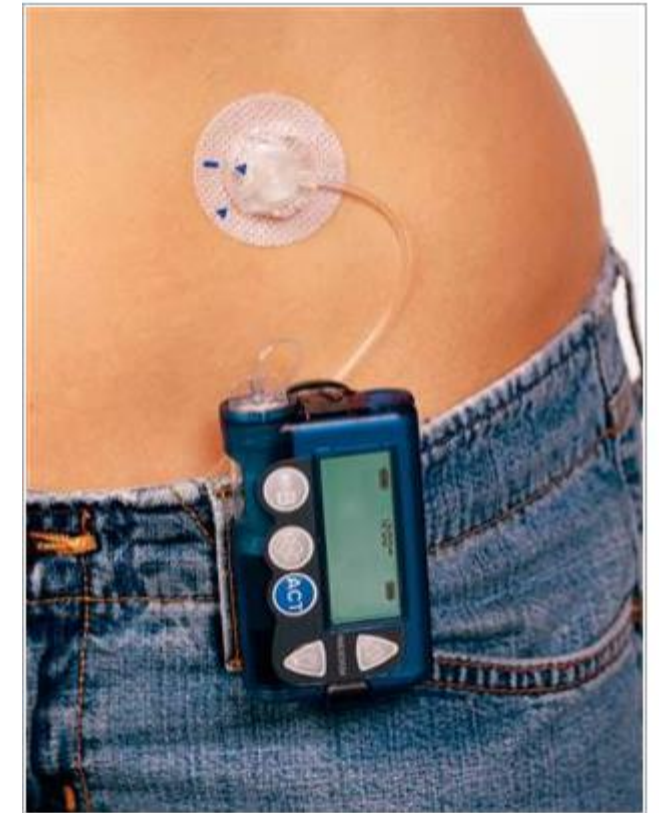
Injectors

Injection Aids

Pen delivery



Insulin Pump



Medtronic Diabetes / AP

What's Wrong with these Values?

TJ is a 52-year-old male with type 1 diabetes and hyperlipidemia presents to the clinic for DM follow-up. He denies of any complaints. Patient is on multiple medications including glargine insulin and lispro insulin before meals. He brings in home glucose readings as shown here. HgA1c 9.1%.

What is the most appropriate recommendation for this patient?

FBG	After Break- fast	Befor e Lunc h	After Lunch	Before Dinne r	After Dinn er	Bed- Time
150	142	90	226	101		206
166		85	244	96	199	
158	132		238			198

What's Wrong with these Values? - Answer

TJ is a 52-year-old male with type 1 diabetes and hyperlipidemia presents to the clinic for DM follow-up. He denies of any complaints. Patient is on multiple medications including glargine insulin and lispro insulin before meals. He brings in home glucose readings as shown here. HgA1c 9.1%.

What is the most appropriate recommendation for this patient?

Answer: increase lispro (bolus insulin) since post meals are elevated. Pre-meals are normal

FBG	After Break-fast	Before Lunch	After Lunch	Before Dinner	After Dinner	Bed-Time
150	142	90	226	101		206
166		85	244	96	199	
158	132		238			198

Group 2 : Insulin Secretagogues

Sulfonylureas (SUs)

Commonly use Agents: Glyburide, Glipizide, Glimepiride

MOA: Enhances Insulin *Secretion* by closing K-ATP channels on Beta cell membranes

Kinetic: eliminated renally

Effect on Weight: **Increased** (~2kg)

HbA1c: up to **1.5%** reduction

Sulfonylureas (SUs)

Major Side Effect: Hypoglycemia

- Esp is 1st and 2nd generation agents (ex: glyburide)
- Hypoglycemia is worse in elderly or with liver/**renal** insufficiency
- Hypoglycemia can be prolonged and severe requiring hospitalization (esp with glyburide!)
- **Less hypoglycemia with glimepiride & glipizide**

Drug	Daily Dose Range (Average)	Elimination Half-life (Hours)	Duration of Action (Hours)	Potency
<u>FIRST GENERATION</u>				
Chlorpropamide	100 mg-500 mg (250 mg)	30-36	2-3 days	+++
Tolazamide	100 mg-1 g (250 mg)	4.7-8	10-14	+++
Tolbutamide	500 mg-2 g (2 g)	4-8	6-12	+
<u>SECOND GENERATION</u>				
Glipizide	5 mg-30 mg (15-20 mg)	2-4	10-24	++++
Glyburide	1.25 mg-20 mg	4.6-12	10-24	++++
Glimepiride	1 mg-8 mg	---	12-24	++++

Meglitinides (secretagogues)

Agents: Repaglinide, Nateglinide

MOA: enhances insulin secretion thru the K-ATP channels

- **Shorter duration of action than sulfonylureas**

Effect on Weight: **Increase**

HbA1c: up to **1.5%** reduction (with repaglinide)

Hypoglycemia: Yes (less than the SU's)

Question - Poll

A 52-year-old man with recently diagnosed type 2 diabetes mellitus comes to the physician for a follow-up examination. Physical examination shows no abnormalities. Laboratory studies show an increased hemoglobin A1c despite patient compliance with diet and exercise recommendations. Treatment with a sulfonylurea is started. Which of the following is most likely to occur in this patient?

- A. Decreased entry of glucose into the muscle cells
- B. Decreased secretion of insulin from the pancreas
- C. Decreased speed of carbohydrate absorption from the intestines
- D. Increased production of glucose from the liver
- E. Increased secretion of insulin from the pancreas

Question – answer

A 52-year-old man with recently diagnosed type 2 diabetes mellitus comes to the physician for a follow-up examination. Physical examination shows no abnormalities. Laboratory studies show an increased hemoglobin A1c despite patient compliance with diet and exercise recommendations. Treatment with a sulfonylurea is started. Which of the following is most likely to occur in this patient?

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- C. Decreased speed of carbohydrate absorption from the intestines
- D. Increased production of glucose from the liver
- E. Increased secretion of insulin from the pancreas**

Group 3 – Drugs that Lower Glucose Levels by Actions on Liver, Muscle, Adipose tissue, and Brain

Thiazolidinediones (‘Glitazones’ or TZDs)

Agent: pioglitazone

MOA:

- *PPAR- γ* activators (peroxisome proliferator-activated receptor gamma) in adipocytes
- Peripheral ‘Insulin sensitizer’: promotes fatty acid uptake peripherally
- *Increase muscle, fat and liver sensitivity to insulin by decreasing free fatty acids levels*

Effect on Weight: Increased (“but it’s good fat”)

HbA1c: up to 0.5-1.4% reduction

Side effects: risk of hypoglycemia, weight gain, **fluid retention, hepatotoxicity**

Glucagon-like peptide-1 (GLP-1) Receptor Agonists and GIP/GLP-1 Dual Agonists*

MOA

- Potentiates glucose-stimulated insulin secretion
- Depresses glucagon secretion
- Delays gastric emptying
- Suppresses appetite

HbA1c: up to lowered by 1.5%

Effect on Weight: **Decrease (3-4kg)***

***Weight loss greater with semaglutide and tirzepatide**

Agents

- Exenatide
- **Liraglutide**
- Lixisenatide
- **Semaglutide**
- **Dulaglutide**

- *Tirzepatide
 - (GLP-1R and GIP-R Dual Agonist)

GLP-1 Receptor Agonists

Side Effects:

- GI most common (nausea, vomiting, diarrhea), **pancreatitis**, cholecystitis
- medullary thyroid cancer / C cell hyperplasia in rats, no reported cases in humans at this time

Contraindicated:

- pancreatitis, family history of medullar thyroid cancer*

Hypoglycemia: **minimal**

*Somewhat controversial, but in the FDA package insert.

HEALTH & WELLNESS

What is 'Ozempic face'? Significant weight loss on the drug can change your face, doctors say

Significant weight loss with drugs like Ozempic and Wegovy, prescribed to treat Type 2 diabetes and obesity, can lead to changes in facial appearance, doctors say.

Profound loss of weight can lead to changes in appearance that some may view as undesirable.

Ozempic brand name for semaglutide



The weight loss is visible on Jeannine DellaVecchia's face. Courtesy Jeannine DellaVecchia

DPP-IV inhibitors

Agents: **Sitagliptin**, **Saxagliptin**, Linagliptin, Alogliptin

MOA: Dipeptidyl peptidase-4 degrades GLP-1 and GIP (incretins),
DPP-4 inhibitors enhance incretin effects

HbA1c: reduced by up to 0.6-0.9%

Weight: neutral

Side Effects: upper respiratory infections, **pancreatitis**, minimal hypoglycemia

Amylin Analog

Agent: Pramlintide (Injectable synthetic analogue of the protein amylin)

MOA: **Slows gastric emptying, inhibits glucagon production, decreases postprandial glucose**

HbA1c: reduced up to 0.5-0.7%

Side Effects: GI, nausea that abates with time, headache

Weight loss: **1-1.5kg**

For use with regular or rapid-acting insulin

Biguanide

Agent: Metformin

MOA:

- Decrease Hepatic Glucose Output
- Decrease Intestinal Glucose Absorption
- Insulin *sensitizer*

Metformin

Effect on weight: **Neutral**

Hypoglycemia: None when used as monotherapy

HbA1c: up to **2.0%** reduction

Elimination: **excreted unchanged in the urine**

Side Effects: **Gastrointestinal, B12 deficiency, lactic acidosis (1/100,000 cases)**

Contraindications:

Renal function considerations:

- **eGFR 30-45 mL/minute/1.73 m²** is *not* recommended (contraindicated <30 mL/minute/1.73 m²)
- Recommended to hold for 48 hours before and after iodinated contrast media if eGFR is 30-60.

Alpha-glucosidase inhibitors

Agents: acarbose, miglitol

MOA: inhibit alpha-glucosidase enzymes that convert complex carbohydrates into simple carbohydrates

HgA1c reduction: 0.3-1%

Side effects: mainly GI related, minimal hypoglycemia

Question - Poll

Drugs from which of the following classes are most likely to promote weight loss?

- A. sulfonylureas
- B. α -glucosidase inhibitors
- C. incretin mimetics
- D. thiazolidinediones
- E. synthetic insulins

Question - answer

Drugs from which of the following classes are most likely to promote weight loss?

- A. sulfonylureas
- B. α -glucosidase inhibitors
- C. incretin mimetics**
- D. thiazolidinediones
- E. synthetic insulins

Case - Poll

64yo M h/o DMII comes in for follow up. He was diagnosed about 3 months ago, he is morbidly obese and having a difficult time losing weight. Since diagnosis you have titrated him up to the maximum recommended dose of metformin. His A1C is still 8.7%, with his logs revealing both elevated fasting and post-prandial sugars. Which of the following is the next best step in management?

- A. rapid acting insulin with meals
- B. stop the metformin
- C. long-acting insulin once daily
- D. acarbose
- E. semaglutide

Case - answer

64yo M h/o DMII comes in for follow up. He was diagnosed about 3 months ago, he is morbidly obese and having a difficult time losing weight. Since diagnosis you have titrated him up to the maximum recommended dose of metformin. His A1C is still 8.7%, with his logs revealing both elevated fasting and post-prandial sugars. Which of the following is the next best step in management?

- A. rapid acting insulin with meals
- B. stop the metformin
- C. long-acting insulin once daily
- D. acarbose
- E. semaglutide**

Question - Poll

A 45-year-old woman presents to your office with a serum glucose of 250 mg/dL and you diagnose diabetes mellitus type II. You intend to prescribe the patient metformin, but you decide to order laboratory tests before proceeding. Which of the following basic metabolic panel values would serve as a contraindication to the use of metformin?

- A. $K^+ > 4.0$
- B. $Na^+ > 140$
- C. $HCO_3^- > 30$
- D. Glucose > 300
- E. eGFR $25 \text{ mL/minute/1.73 m}^2$

Question – answer

A 45-year-old woman presents to your office with a serum glucose of 250 mg/dL and you diagnose diabetes mellitus type II. You intend to prescribe the patient metformin, but you decide to order laboratory tests before proceeding. Which of the following basic metabolic panel values would serve as a contraindication to the use of metformin?

- A. $K^+ > 4.0$
- B. $Na^+ > 140$
- C. $HCO_3^- > 30$
- D. Glucose > 300
- E. **eGFR 25 mL/minute/1.73 m²**

SGLT2 inhibitors

Agents: Canagliflozin, **Dapagliflozin**, **Empagliflozin**, Ertugliflozin, Bexagliflozin

MOA: SGLT2 (sodium glucose transporter) responsible in reabsorption of glucose in proximal tubule of the kidney. SGLT2 inhibitors **reduce glucose reabsorption, thus increasing glycosuria**

Effect on weight: 2- 3Kg reduction

Hba1c: 0.5-1% reduction

Adverse reactions:

- **Hypovolemia**, hyperkalemia
- **Genital mycotic and urinary tract infections** (3 -fold risk)
- Acute kidney injury
- Euglycemia ketoacidosis

Question - Poll

In monotherapy, medication from which of the following classes is most likely to cause hypoglycemia as an adverse effect?

- A. sulfonylurea
- B. SGLT2 inhibitor
- C. thiazolidinedione
- D. α -glucosidase inhibitor
- E. biguanide

Question – answer

In monotherapy, medication from which of the following classes is most likely to cause hypoglycemia as an adverse effect?

- A. **sulfonylurea**
- B. SGLT2 inhibitor
- C. thiazolidinedione
- D. α -glucosidase inhibitor
- E. biguanide

Question – poll

A patient with history of long-standing type 2 DM, hypertension, depression, and gastroparesis. Her current A1c is 8.2% and she is unable to tolerate metformin due to gastrointestinal side effects. Which of the following agent is NOT the most appropriate option for this patient's diabetes care?

- A. liraglutide
- B. glipizide
- C. pioglitazone
- D. canagliflozin

Question – answer

A patient with history of long-standing type 2 DM, hypertension, depression, and gastroparesis. Her current A1c is 8.2% and she is unable to tolerate metformin due to gastrointestinal side effects. Which of the following agent is NOT the most appropriate option for this patient's diabetes care?

- A. liraglutide**
- B. glipizide
- C. pioglitazone
- D. canagliflozin

Question - Poll

19yo F with DM I on insulin presents to your clinic for routine follow up. She was diagnosed about 3 months ago, and she is currently taking insulin glargine 15 units at nighttime, and insulin aspart 6 units with breakfast, 6 units with lunch and 8 units with dinner. She denies any sweats, confusion, jitteriness. She brings her meter with her today, which shows fasting AM blood sugars of 170-190, with post-prandial sugars of 120-150. How would you like to change her regimen?

- A. Increase her breakfast aspart to 8, lunch to 8 and dinner to 10 units
- B. Start her on metformin
- C. Start her on glipizide
- D. Increase her lunch aspart to 12 units
- E. Increase her glargine to 18 units

Question - Poll

19yo F with DM I on insulin presents to your clinic for routine follow up. She was diagnosed about 3 months ago, and she is currently taking insulin glargine 15 units at nighttime, and insulin aspart 6 units with breakfast, 6 units with lunch and 8 units with dinner. She denies any sweats, confusion, jitteriness. She brings her meter with her today, which shows fasting AM blood sugars of 170-190, with post-prandial sugars of 120-150. How would you like to change her regimen?

- A. Increase her breakfast aspart to 8, lunch to 8 and dinner to 10 units
- B. Start her on metformin
- C. Start her on glipizide
- D. Increase her lunch aspart to 12 units
- E. Increase her glargine to 18 units**

Relationship of average Blood Glucose to HbA1c

The HbA1c test is a measure of the average long term Blood Glucose level because RBC's tend to remain in circulation for 100 to 120 days

HgbA1c	Average Blood Sugar		Interpretation
4		61	5.6 and less -Normal
5		92	
6		124	5.7-6.5 - Pre-diabetic
7		156	6.6 and above – Diabetic Diabetic patients under treatment should ideally have a HgbA1c less than 7.0
8		188	
9		219	
10		251	
11		283	
12		314	

Symptoms of moderate to severe hypoglycemia?

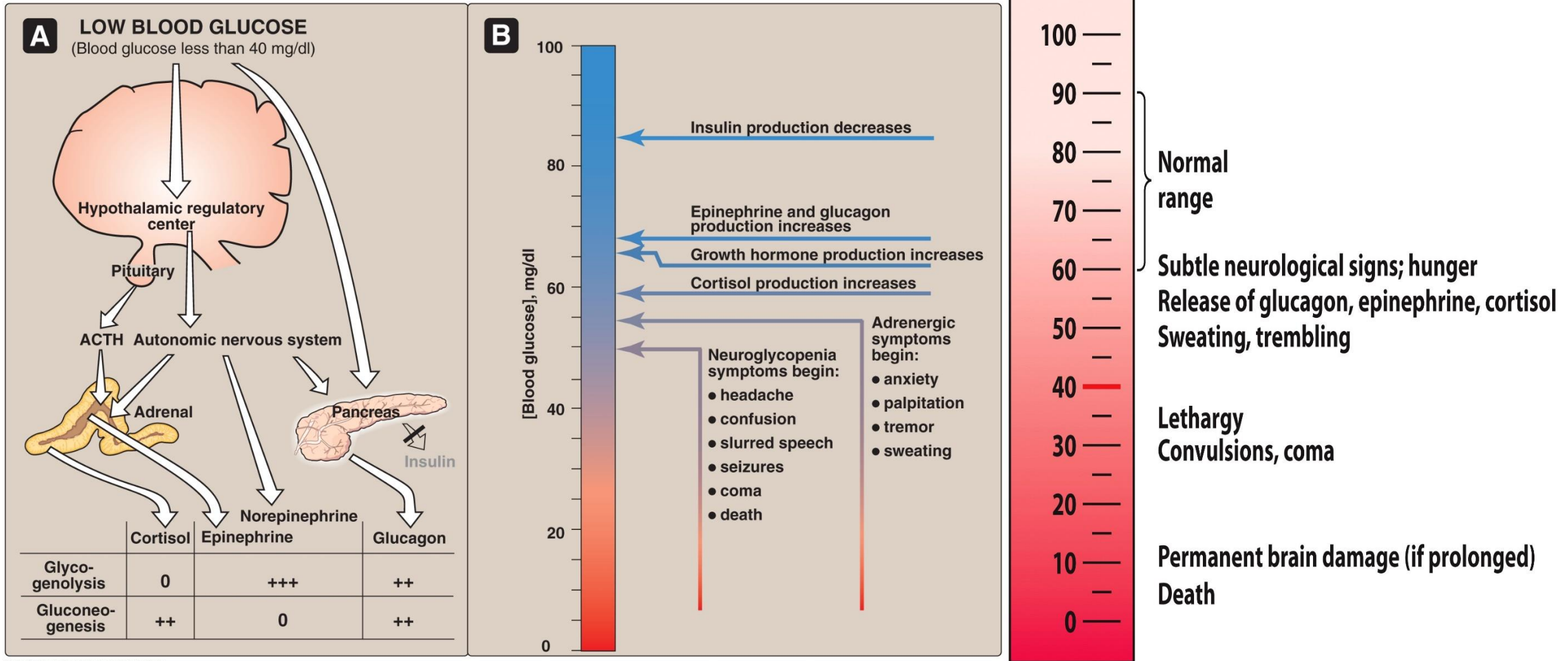


Figure 23-24

Lehninger Principles of Biochemistry, Sixth Edition

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Classification of Hypoglycemia

Level 1: A glucose alert value of 70 mg/dL or less.

Level 2: A glucose level of 54 mg/dL is sufficiently low to indicate potentially serious, clinically important hypoglycemia.

Level 3: Severe hypoglycemia, as defined by the ADA, denotes severe cognitive impairment requiring external assistance for recovery.

Ibrahim M, Baker J, Cahn A, et al. Hypoglycemia and its management in primary care setting [published online ahead of print, 2020 Apr 28]. *Diabetes Metab Res Rev.* 2020;e3332. doi:10.1002/dmrr.3332

Hypoglycemia and Intervention

Hypoglycemia

- Common complication
- Major limiting factors to optimal glycemic control

Intervention

- “rule of 15”
 - 15 gram of fast-acting carbohydrates (1/2 cup of milk/juice/soda)
 - Check glucose in 15 minutes, if glucose below 70mg/dl, repeat
 - If glucose is above 70mg/dl, eat a normal snack or meal

Glucagon



Indicated as a treatment for severe hypoglycemia, which may occur in patients with diabetes mellitus.

Supplementary carbohydrate should be given as soon as possible, especially in pediatric patients.

Case

A 67 y/o woman with uncontrolled type 2 DM despite maximal dosage of glucophage, pioglitazone, and insulin 70/30 35units qam, 28 units qpm, presents to the clinic and is wondering if any herbal supplement that will help control her diabetes?

HgA1C is 9.2 today

Herbal Supplements (extra information)

Ginseng

Ginsenosides

- Chemical & potency varies
- CNS stimulant
- Alters lipid, carbohydrate metabolism
- Stimulates the immune system
- Interferes with the platelet aggregation

Indications

- Tonic, stimulant, diabetes, impotence, insomnia, stress resistance

Fenugreek Seed

High in fiber

- trigonelline, nicotinic acid, and coumarin
- Improves plasma glucose & insulin response
- reduces postprandial glucose levels
- cardiotonic, diuretic, anti-inflammatory, antihypertensive effects

Dosage: 15-40mg/day

Adverse rxn:

- allergic rxns, diarrhea, flatulence

Garlic / Fiber

Garlic

- reduces blood glucose & increases insulin in healthy volunteers

Lifestyle modification

- High fiber, low fat, high vegetable
- 35-40 gram / day
- HgA1c ↓ by >2 %

Other Herbal Supplements

Bitter melon

- proposed mechanism
 - direct action similar to insulin
 - inhibits gluconeogenesis
 - 15 to 50mg per day

Cinnamon

3-6 months trials, participants
60-100/trial
3 to 5 g/day
Delayed gastric emptying

Gymnema

- native plant of India “sugar blocker”
 - enhances endogenous insulin production
 - 400mg/day extract
 - reduces taste sensation of sweetness & bitterness

Remember...

- Individualize treatment and glycemic goals to your patient
- Always emphasize lifestyle changes
- Review diabetic diet, low fat diet
- Assess for medication adherence
- Encourage weight loss and regular exercise (there is always some form exercise for everyone)

Key Points: Drugs for T2DM

- Metformin is the typical initial agent, provided there is no contraindication for its use or compelling reasons for an alternative agent
- Multiple classes of agents are available as add-on agents or if metformin is not an option
- Each class of anti-diabetes medications has its own good and bad features
- Agents such as metformin, GLP-1R agonists, DPP-4 inhibitors, and SGLT2 inhibitors are weight neutral or have potential for weight loss

Summary of Agents

Drug	Major Side Effects
Insulin	Acute hypoglycemia, weight gain
Sulfonylurea	Acute hypoglycemia, weight gain
TZD	Weight gain, edema, risk of bladder cancer (pioglitazone), hepatic failure
GLP-1R & GLP-1R/GIP-R agonists	Pancreatitis, cholelithiasis, theoretical risk of medullary thyroid cancer
DPP-4 inhibitor	Pancreatitis
Amylin analog	N/V, headache
Metformin	Lactic acidosis, N/V/D, B12 deficiency. Avoid in severe renal impairment.
SGLT2 Inhibitor	Increased urination, weight loss, hypotension, genital mycotic infection, UTI. Avoid in severe renal impairment.
Alpha-glucosidase inhibitor	GI